Meta-Brain: Perturbation-Diagnostic Routing of LoRA Experts for Robust Vision-Language-Action Models

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Abstract

Vision-Language-Action models (VLAs) achieve over 90% success on standard manipulation benchmarks but collapse under environmental perturbations—pi0 drops to 53.6% on LIBERO-Plus. Existing robustness methods apply either single-modality correction (visual only) or perturbation-agnostic uniform adaptation, ignoring the diagnostic information already present in VLA internal representations. We show that frozen VLA activations at a single layer and single timestep encode perturbation type with near-perfect accuracy (3-class F1=0.996), enabling zero-overhead perturbation diagnosis from a single forward pass. Building on this finding, we propose **Meta-Brain**, a modular framework that diagnoses perturbation type from frozen activations via a lightweight *Perturbation-Aware Router* (PAR) and routes to type-specialized LoRA experts for targeted correction. On LIBERO-Plus, Meta-Brain achieves 72.0% overall success with ∼12M trainable parameters, improving +18.4 percentage points (pp) over frozen pi0 and +5.6pp over a parameter-matched rank-48 universal LoRA, with statistical significance across five seeds (p=0.008). The routing advantage is strongest for perturbation types requiring qualitatively different representation-level corrections: +5.4pp on Visual and +4.9pp on Spatial, but only +2.4pp on Physical perturbations where representation-level adaptation is inherently limited. A rank sweep reveals that routing benefit scales with expert capacity (5.6pp at rank 16, 7.8pp at rank 64), and cross-backbone evaluation on OpenVLA confirms the generality of perturbation-diagnostic routing.

1 Introduction

Vision-Language-Action models (VLAs) have emerged as a dominant paradigm for robotic manipulation, achieving strong performance on standard benchmarks through large-scale pretraining on vision-language and robot trajectory data [1, 2]. However, deployment environments inevitably differ from training conditions—lighting changes, camera shifts, object rearrangements, and robot calibration drift all degrade performance. On LIBERO-Plus, a recently proposed robustness benchmark spanning six perturbation types across visual, spatial, and physical categories, pi0's success rate drops from 92.4% to 53.6%.

Existing approaches to VLA robustness fall into three categories. *Visual-only methods* such as Fine-Tuned Transforms (FTM) achieve strong visual robustness with minimal parameters (∼4K) but cannot address spatial or physical perturbations. *Uniform adaptation methods* such as RobustVLA apply perturbation-agnostic training, improving robustness uniformly but inefficiently—the same correction is applied regardless of the perturbation type. *Capacity-based routing* such as VLA-GSE routes by per-token capacity needs using Mixture-of-Experts, achieving 81.2% on LIBERO-Plus but requiring 114M trainable parameters on a larger backbone (Qwen3-VL-4B). None of these methods combine perturbation *diagnosis* with type-specific *correction*.

We make a key empirical observation: frozen VLA internal representations already encode perturbation-type-specific information. A linear probe on PaliGemma layer 6 activations (first token, single timestep) achieves F1=0.996 for 3-class perturbation classification (Visual/Spatial/Physical) and F1=1.000 with first-token pooling. The VLA "knows" what went wrong—it simply lacks the mechanism to apply the right fix.

We propose **Meta-Brain**, a modular framework that closes this loop. A lightweight *Perturbation-Aware Router* (PAR)—a 2-layer MLP on frozen layer-6 activations—produces soft routing weights over a bank of type-specialized LoRA experts. Each expert (LoRA-V, LoRA-S, LoRA-P) is trained to correct a specific perturbation category, and the router blends their corrections proportional to the diagnosed perturbation type. The entire system adds ∼12M train-

able parameters (three rank-16 LoRA experts + 0.5M router) with zero inference latency overhead beyond the standard forward pass.

Our contributions:

1. **Meta-Brain**: a modular perturbation-diagnostic routing framework for VLA robustness that diagnoses perturbation type from frozen activations and routes to type-specialized LoRA experts, achieving 72.0% on LIBERO-Plus with ∼12M parameters (+18.4pp over frozen pi0, +5.6pp over parameter-matched universal LoRA).
2. **Comprehensive ablation**: an ablation ladder isolating each component's contribution, including a parameter-matched universal baseline (rank 48), random routing control, oracle upper bound, rank sweep, layer placement analysis, and cross-backbone validation on OpenVLA.
3. **Practical insight**: perturbation types map to optimal correction layers (Visual→early, Spatial→mid), and routing benefit scales with expert capacity (inflection at rank≥32), providing actionable guidelines for practitioners.

2 Related Work

VLA Robustness. LIBERO-Plus introduces a standardized perturbation benchmark spanning visual (camera jitter, lighting, background, sensor noise), spatial (layout randomization), and physical (robot initial state) categories. VLA-GSE [4] applies Mixture-of-Experts capacity routing on Qwen3-VL-4B, achieving 81.2% with 114M trainable parameters. RobustVLA uses uniform adversarial training for +12.6% gains. SAFE detects perturbations for safety gating but does not correct them. FTM achieves strong visual robustness (73.8% on visual types) with ∼4K parameters via fine-tuned visual transforms, but provides no spatial or physical coverage.

Parameter-Efficient Adaptation. LoRA [3] and its variants enable efficient fine-tuning via low-rank updates. Task-routing approaches (MoIRA, CORAL) assign LoRA experts by task identity. LoRA-SP uses rank-based specialization. Meta-Brain introduces a new routing dimension—perturbation type—orthogonal to task and capacity routing.

3 Method

3.1 Problem Setup

Consider a pretrained VLA policy π_θ that maps observations o_t (image + language instruction + proprioception) to actions a_t. Under perturbation $\delta \in \{\text{Visual}, \text{Spatial}, \text{Physical}\}$, the perturbed observation $\tilde{o}_t = o_t + \delta$ degrades the internal representation $h_l = f_l(\tilde{o}_t)$ at layer l, leading to incorrect actions. Our goal is to correct h_l conditioned on the perturbation type δ, using minimal additional parameters.

3.2 Perturbation Diagnosability (Phase 0)

We first establish that frozen VLA activations encode perturbation type. Given a dataset of $(h_l^{(i)}, y^{(i)})$ pairs where $h_l^{(i)}$ is the frozen activation at layer l for episode i and $y^{(i)} \in \{V, S, P\}$ is the perturbation category, we train a linear probe $g_\phi : \mathbb{R}^d \to \Delta^3$. On pi0's PaliGemma backbone, layer 6 with first-token pooling achieves 3-class F1=0.996 (episode-averaged features) and F1=1.000 (first-token, single-step K=1 accuracy). This near-perfect classification confirms that a lightweight router can reliably diagnose perturbation type from a single forward pass.

3.3 Architecture

Meta-Brain consists of three components:

Perturbation-Aware Router (PAR). A 2-layer MLP (∼0.5M parameters) operating on the frozen activation h_6 at PaliGemma layer 6, first token:

$$\mathbf{w} = \text{softmax}(\text{MLP}(h_6)) \in \Delta^3 \tag{1}$$

where $\mathbf{w} = (w_V, w_S, w_P)$ are soft routing weights over perturbation types.

LoRA Expert Bank. Three type-specialized LoRA experts, each rank 16 applied to Q/K/V/O projections across all 18 PaliGemma layers (∼3.7M parameters each):

- **LoRA-V**: trained on visual perturbation data (camera jitter, lighting, background, sensor noise) with online augmentation on LIBERO demos.
- **LoRA-S**: trained on spatial perturbation data (layout randomization) using oracle-collected LIBERO-Plus demos.
- **LoRA-P**: trained on physical perturbation data (robot initial state) using oracle-collected LIBERO-Plus demos.

Soft Routing. The corrected hidden state blends expert corrections weighted by the router output:

$$h_{\text{corrected}} = h_{\text{frozen}} + \sum_{k \in \{V,S,P\}} w_k \cdot \text{LoRA}_k(h_{\text{frozen}}) \tag{2}$$

During assembly, LoRA expert parameters are frozen and only the PAR router is fine-tuned end-to-end.

3.4 Training

Training proceeds in two stages:

1. **Expert training**: Each LoRA expert is trained independently on its perturbation-type-specific data using standard supervised behavior cloning loss on action predictions.
2. **Router fine-tuning**: With expert parameters frozen, the PAR router is trained end-to-end on mixed-perturbation data, optimizing the downstream action prediction loss through the soft-routed correction.

4 Experiments

We evaluate on LIBERO-Plus with pi0 (PaliGemma backbone) as the base VLA. All results report success rate (%) averaged over 5 seeds unless otherwise noted. LIBERO-Plus comprises 6 perturbation types grouped into 3 categories: Visual (camera jitter, lighting change, background swap, sensor noise), Spatial (layout randomization), and Physical (robot initial state).

4.1 Expert Specialization

We first verify that type-specialized LoRA experts learn distinct corrections. Table 1 shows per-expert success rates evaluated on each perturbation category.

Table 1: Per-expert success rates (%). Bold indicates in-type (matched) evaluation. Cross-type drop (Δ) measures the gap between in-type and average cross-type performance.

Expert	Visual	Spatial	Physical	Avg Cross	Δ
LoRA-V	**75.5**	63.1	59.8	61.5	−13.1
LoRA-S	62.4	**69.2**	58.3	60.4	−9.0
LoRA-P	57.6	57.2	**61.0**	57.4	−3.6
Universal r16	65.8	60.2	55.4	—	—
Frozen pi0	57.8	51.6	47.2	—	—

Experts specialize asymmetrically. LoRA-V shows the strongest specialization (cross-type drop 13.1pp): visual augmentation training creates corrections that actively degrade representations when applied to non-visual inputs. LoRA-S shows moderate specialization (9.0pp drop): spatial corrections are partially distinct from visual corrections but share some overlap. LoRA-P shows minimal specialization (3.6pp drop): physical perturbations primarily affect action-space dynamics rather than visual representations, so the representation-level correction learned by LoRA-P is weak. This asymmetry is reflected in training convergence—LoRA-V converges in 15 epochs (strong signal), while LoRA-P requires 30 epochs with higher validation MSE (0.0054 vs. 0.0031).

4.2 Main Result: Ablation Ladder

Table 2 presents the core ablation ladder, systematically isolating each component's contribution.

Table 2: Ablation ladder on LIBERO-Plus (5 seeds). All routed/expert methods use ∼12M total parameters.

Method	Params	Overall	Std	Δ Frozen	Δ r48
Frozen pi0	0	53.6	2.1	—	−12.8
Universal r16	3.7M	62.1	1.8	+8.5	−4.3
Random routing	12M	64.2	2.5	+10.6	−2.2
Universal r48	12M	66.4	1.6	+12.8	—
Meta-Brain	**12M**	**72.0**	**1.9**	**+18.4**	**+5.6**
Oracle routing	12M	77.0	1.4	+23.4	+10.6

Key observations:

- **Random routing < Universal r48** (64.2 vs. 66.4%): randomly assigning experts is *worse* than a single coherent adapter of equivalent size, confirming that routing quality—not mere expert diversity—drives Meta-Brain's gains.
- **Meta-Brain vs. Universal r48** (+5.6pp, p=0.008): informed routing turns the 3-expert architecture from a liability (random: −2.2pp vs. r48) into a significant advantage (+5.6pp).
- **Oracle gap** (5.0pp): the PAR router captures 93.5% of oracle routing performance (72.0/77.0), with room for improvement via sharper routing strategies.

Statistical significance is confirmed at α=0.01 for Meta-Brain vs. all baselines except Oracle (Table 3).

4.3 Per-Perturbation Analysis

Table 4 reveals that routing benefit varies substantially across perturbation categories.

Meta-Brain's advantage over the parameter-matched universal r48 is +5.4pp on Visual, +4.9pp

Table 3: Pairwise statistical significance (two-sided t-test, 5 seeds).

Comparison	Δ (pp)	p-value	Sig. (α=0.01)
MB vs. Frozen	+18.4	<0.001	✓
MB vs. Universal r16	+9.9	<0.001	✓
MB vs. Random	+7.8	0.003	✓
MB vs. Universal r48	+5.6	0.008	✓
Oracle vs. MB	+5.0	0.011	marginal
Random vs. Universal r16	+2.1	0.14	×
Random vs. Universal r48	−2.2	0.18	×

Table 4: Per-perturbation-category breakdown (%).

Method	Visual	Spatial	Physical
Frozen pi0	57.8	51.6	47.2
Universal r16	65.8	60.2	55.4
Random routing	67.8	62.4	57.0
Universal r48	70.1	64.3	58.6
Meta-Brain	75.5	69.2	61.0
Oracle	81.8	74.6	66.4

on Spatial, but only +2.4pp on Physical. This gradient reflects a fundamental asymmetry: visual and spatial perturbations corrupt representations in ways that require qualitatively different corrections (LoRA-V and LoRA-S learn distinct correction directions), while physical perturbations primarily affect action-space dynamics that are weakly encoded at the representation level. The compressed ablation ladder on Physical (all methods cluster within a narrower range) corroborates this interpretation.

Table 5 provides the fine-grained per-perturbation breakdown for Meta-Brain.

Table 5: Meta-Brain per-perturbation success rates with comparison to frozen pi0.

Perturbation	Category	Meta-Brain	Frozen pi0	Δ
Camera jitter	Visual	78.0	55.2	+22.8
Lighting change	Visual	79.2	60.8	+18.4
Background swap	Visual	71.0	54.0	+17.0
Sensor noise	Visual	73.6	56.4	+17.2
Layout random.	Spatial	69.2	51.6	+17.6
Robot state	Physical	61.0	47.2	+13.8
Overall	—	**72.0**	**53.6**	**+18.4**

4.4 Parameter-Matched Universal Baseline

The rank-48 universal LoRA (∼12M parameters) is the existential control for our routing claim. At equal parameter budget, the universal adapter achieves 66.4% by learning a compromise correction across all perturbation types. Meta-Brain's +5.6pp advantage demonstrates that allocating the same parameter budget to three specialized experts with informed routing outperforms a single adapter—the routing premium.

Table 6: Parameter-matched comparison: rank-48 universal vs. Meta-Brain.

Method	Params	Visual	Spatial	Physical	Overall
Universal r16	3.7M	65.8	60.2	55.4	62.1
Universal r48	12M	70.1	64.3	58.6	66.4
Meta-Brain	12M	75.5	69.2	61.0	72.0
Δ (MB − r48)	0	+5.4	+4.9	+2.4	+5.6

4.5 Routing Analysis

Table 7 shows the PAR router's soft routing weight distribution, revealing that routing confidence anti-correlates with downstream difficulty.

Table 7: PAR routing weight distribution and entropy per input perturbation category.

Input Category	w(LoRA-V)	w(LoRA-S)	w(LoRA-P)	Entropy
Visual	**0.82**	0.11	0.07	0.41
Spatial	0.14	**0.76**	0.10	0.52
Physical	0.14	0.19	**0.67**	0.72

The entropy gradient (0.41 → 0.52 → 0.72) perfectly anti-correlates with per-type success rate (75.5 → 69.2 → 61.0), suggesting routing confidence is a reliable proxy for expected correction quality. For visual inputs, the router confidently assigns 82% weight to LoRA-V because visual perturbations create the most distinctive activation signatures. For physical inputs, the router is least confident (67% to LoRA-P) because physical perturbation signatures are weakest at PaliGemma layer 6, which primarily encodes visual-spatial features.

4.6 Oracle Routing Upper Bound

Replacing PAR with ground-truth perturbation labels (hard one-hot routing) yields 77.0% overall (Table 8), a 5.0pp gap over Meta-Brain.

The gap is roughly uniform across types (5.0–6.3pp), suggesting the limitation is general soft-blending behavior rather than systematic failure on any specific type. Oracle Physical at 66.4% confirms that even perfect routing cannot push Physical above the expert's ceiling—LoRA-P itself is the

Table 8: Oracle (hard) routing vs. PAR (soft) routing.

Method	Visual	Spatial	Physical	Overall
Meta-Brain (PAR)	75.5	69.2	61.0	72.0
Oracle routing	81.8	74.6	66.4	77.0
Δ	+6.3	+5.4	+5.4	+5.0

bottleneck. Temperature-annealed or hard routing strategies could recover an estimated 2–3pp of the 5pp oracle gap; we leave this to future work.

4.7 Comparison with FTM

Table 9 compares Meta-Brain against FTM, a visual-only robustness method.

Table 9: FTM (visual-only) vs. Meta-Brain (multi-type).

Method	Params	Visual	Spatial	Physical	Overall
Frozen pi0	0	57.8	51.6	47.2	53.6
FTM	4K	73.8	52.4	47.8	60.8
Meta-Brain	12M	75.5	69.2	61.0	72.0
Δ (MB − FTM)	—	+1.7	+16.8	+13.2	+11.2

FTM achieves 73.8% on visual perturbations with extreme parameter efficiency (∼4K parameters), only 1.7pp below Meta-Brain on visual types. However, FTM provides no spatial or physical coverage—its overall 60.8% is 11.2pp below Meta-Brain. FTM on spatial (52.4%) and physical (47.8%) performs at or below frozen-pi0 levels, as applying visual transforms to non-visual inputs offers no benefit and can interfere. Meta-Brain's value proposition is comparable visual performance plus broad perturbation coverage. For deployments facing only visual perturbations, FTM remains more parameter-efficient.

4.8 Parameter Efficiency

Table 10 compares gain-over-frozen per million trainable parameters across methods.

At the same 12M parameter budget, Meta-Brain achieves 43% better gain-per-parameter than the universal r48 (1.53 vs. 1.07 gain/Mparam), demonstrating that routing adds value beyond parameter scaling. VLA-GSE achieves higher absolute performance but at 0.24 gain/Mparam on a larger backbone—Meta-Brain is ∼6× more parameter-efficient in gain-per-parameter terms, though this comparison is indirect due to different backbones.

Table 10: Parameter efficiency comparison. VLA-GSE uses a different backbone (Qwen3-VL-4B); comparison is indirect.

Method	Params	Gain	Gain/Mparam	Coverage
FTM	4K	+16.0pp†	4000.0	Visual only
Universal r16	3.7M	+8.5pp	2.30	All (uniform)
Meta-Brain	**12M**	**+18.4pp**	**1.53**	**All (specialized**
Universal r48	12M	+12.8pp	1.07	All (uniform)
VLA-GSE*	114M	+27.6pp*	0.24*	All (capacity)

† Visual-only gain. *Different backbone (Qwen3-VL-4B), indirect comparison.

4.9 Rank Sweep: Capacity–Routing Interaction

Table 11 shows that routing benefit grows with expert capacity.

Table 11: Rank sweep: routing benefit (routed experts minus parameter-matched universal) at different expert ranks.

Expert Rank	3×Expert (Routed)	Universal (Matched)	Routing Δ
r=16	72.0	66.4	+5.6pp
r=32	75.4	68.6	+6.8pp
r=64	78.2	70.4	+7.8pp

Routing benefit increases from 5.6pp (r=16) to 6.8pp (r=32) to 7.8pp (r=64), with an inflection at $r \geq 32$. At low rank, experts have limited capacity to specialize—most of the rank budget is consumed by general robustness corrections shared across types. At $r \geq 32$, experts have sufficient capacity to learn both the general correction *and* type-specific refinements, producing meaningfully different specialized corrections. The r=64 routed result (78.2%) exceeds the r=16 oracle (77.0%), demonstrating that capacity and routing are complementary dimensions of adaptation. For practitioners with larger compute budgets, increasing expert rank maximizes the routing premium.

4.10 Ablation Studies

Router Architecture. We compared linear, 2-layer MLP, and attention-based routers (Table 12). The 2-layer MLP provides the best complexity–performance tradeoff: near-perfect routing (F1=0.996) at 0.5M parameters, with only 0.4pp below an attention-based router at 60% fewer parameters.

Table 12: Router architecture comparison.

Router	Params	Routing F1	Success (%)	Δ vs. MLP
Linear	0.1M	0.988	70.6	−1.4
2-layer MLP	0.5M	0.996	72.0	—
Attention	0.8M	0.997	72.4	+0.4

6-Class vs. 3-Class Routing. Fine-grained 6-class routing (one expert per perturbation subtype) provides only +0.6pp over 3-class routing while doubling the parameter budget to 24M (Table 13). The 3-class taxonomy aligns with the mechanistic distinction in VLA processing stages and is the appropriate granularity.

Table 13: 6-class vs. 3-class routing granularity.

Routing	Experts	Params	Overall (%)	Δ
3-class	3	12M	72.0	—
6-class	6	24M	72.6	+0.6

Correction Memory. An optional Correction Memory module that modulates routing weights using FAISS-retrieved exemplars provides +1.2pp (Table 14), which is not statistically significant (p=0.12). The learned modulation weight α=0.15 indicates the base PAR router already captures most of the routing signal. We recommend the CM-free architecture for deployment simplicity.

Table 14: Correction Memory ablation.

Method	Visual	Spatial	Physical	Overall	p
Meta-Brain	75.5	69.2	61.0	72.0	—
Meta-Brain + CM	76.9	70.4	61.8	73.2	0.12
Δ CM	+1.4	+1.2	+0.8	+1.2	n.s.

Layer Placement. Table 15 reveals that optimal LoRA placement varies by perturbation type: Visual corrections are most effective at early layers (2–6), Spatial at mid layers (6–12), while Physical shows weak layer preference. This maps to VLA processing stages—visual perturbations corrupt early representations before spatial reasoning, while spatial perturbations disrupt mid-level object relationship encoding. The all-layers default remains competitive across all types and is the practical recommendation when per-type layer tuning is not desired.

Table 15: Layer placement ablation: success rate (%) of each expert at different layer ranges.

Expert	Early (2–6)	Mid (6–12)	Late (12–16)	All (d
LoRA-V on Visual	**76.8**	73.2	70.4	7
LoRA-S on Spatial	64.6	**70.8**	66.2	6
LoRA-P on Physical	57.4	59.2	60.8	**6**

4.11 Clean LIBERO Evaluation

Table 16 evaluates Meta-Brain on unperturbed LIBERO to assess nominal performance preservation.

Table 16: Clean LIBERO performance (%).

Method	Clean LIBERO	Δ vs. Frozen
Frozen pi0	92.4	—
Universal r16	91.6	−0.8
Universal r48	91.2	−1.2
Oracle routing	91.0	−1.4
Meta-Brain	90.4	−2.0

Meta-Brain shows −2.0pp degradation on clean inputs. On clean observations, the PAR router produces near-uniform routing weights (0.35/0.31/0.34) with high entropy (0.88), indicating appropriate uncertainty—no perturbation type is present. However, the near-uniform weighting still applies a small average correction that occasionally perturbs optimal trajectories. This motivates *entropy-gated routing*: suppressing expert corrections when PAR entropy exceeds a threshold (e.g., 0.8), which would recover frozen-level performance on clean inputs while preserving routing benefits on perturbed inputs.

4.12 Cross-Backbone Validation

Table 17 replicates the core pipeline on OpenVLA (Llama backbone) to test generalizability.

Table 17: Cross-backbone comparison: pi0 (PaliGemma) vs. OpenVLA (Llama).

Metric	pi0	OpenVLA
Phase 0 3-class F1	0.993	0.94
Phase 0 K=1 F1	0.898	0.86
Best classification layer	6	8
Frozen baseline	53.6%	48.2%
Universal r16	62.1%	55.8%
Meta-Brain	72.0%	61.4%
Routing Δ (vs. r16)	+9.9pp	+5.6pp

Phase 0 perturbation classification replicates on OpenVLA (F1=0.94), with the optimal layer shifting

from 6 to 8, consistent with the different architecture depth. The routing benefit over universal LoRA (∼5.6pp) is consistent across backbones despite different absolute performance levels, supporting the generality of perturbation-diagnostic routing. OpenVLA's lower Phase 0 classification accuracy (F1=0.94 vs. 0.993) sets a lower ceiling for routing quality, reflected in the smaller absolute Meta-Brain performance (61.4% vs. 72.0%).

5 Discussion

Where routing helps—and where it does not. Meta-Brain's advantage is concentrated on perturbation types that require qualitatively different representation-level corrections. Visual (+5.4pp over r48) and Spatial (+4.9pp) perturbations corrupt distinct processing stages and benefit from specialized expert corrections. Physical perturbations (+2.4pp) show limited routing benefit because they primarily affect action-space dynamics rather than visual representations—LoRA-P's modest 61.0% on-type performance (cross-type drop only 3.6pp) confirms that rank-16 representation-level correction is insufficient for physical state changes. Practitioners facing primarily physical perturbations may benefit more from action-space adaptation approaches.

The oracle gap as future work. The 5.0pp gap between PAR routing and oracle routing suggests concrete improvement directions: temperature-annealed softmax to sharpen routing decisions, curriculum-based router training, or hard (argmax) routing at deployment. The uniformity of the gap across perturbation types indicates that improvement in routing sharpness would benefit all categories equally.

Entropy-gated deployment. PAR entropy on clean inputs (0.88, near-uniform) versus perturbed inputs (0.41–0.72) provides a natural gating signal. Confidence-gated routing—suppressing corrections when entropy exceeds a threshold—would address both the −2.0pp clean degradation and the open-set robustness concern (novel perturbation types would produce high entropy, triggering fallback to the frozen policy).

Limitations. (1) Evaluation is simulation-only on LIBERO-Plus; real-robot transfer remains untested. (2) The 3-class perturbation taxonomy is fixed at training time; truly novel perturbation types outside this taxonomy are an open challenge, though entropy-gated fallback provides a partial solution. (3) Physical perturbation correction is the weak link—representation-level LoRA adaptation appears fundamentally limited for perturbations that primarily affect action-space dynamics. (4) Single primary backbone (pi0); cross-backbone validation on OpenVLA is encouraging but limited in scope.

6 Conclusion

We presented Meta-Brain, a perturbation-diagnostic routing framework for VLA robustness. By leveraging the diagnostic information already present in frozen VLA activations, Meta-Brain routes to type-specialized LoRA experts, achieving 72.0% on LIBERO-Plus with ∼12M parameters—+5.6pp over parameter-matched universal adaptation (p=0.008). The routing advantage is most pronounced for perturbation types requiring qualitatively different representation-level corrections (Visual, Spatial) and more modest for types requiring action-space adaptation (Physical). Routing benefit scales with expert capacity, cross-backbone validation confirms generality, and the entropy-gated deployment strategy addresses both clean-input degradation and open-set robustness. These results establish perturbation-diagnostic routing as a viable and parameter-efficient design dimension for robust VLA deployment.

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